Paleomagnetism of the Narrabeen Chocolate Shales and the Tasmanian Dolerite

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Abstract. The results of field and laboratory studies show that the stable component of the natural remanent magnetization of the Chocolate shales of the Narrabeen series (Lower Triassic) and the Tasmanian dolerite intrusives (Jurassic) have similar almost vertical directions.

Chocolate shales. Results were obtained from the Chocolate shales in the Lower Triassic Narrabeen series [*David*, 1950, p. 409]. The shales are redistributed tuffs with much clastic material; they are found in bands usually not more than a few meters thick. Oriented samples were collected from four sites (CS1 to CS4) at which the stratigraphic thicknesses sampled were 4, 2, 6, and 3 meters, respectively. The sites are separated by distances of about 80 km. The total stratigraphic thickness covered is not known, but it is not less than 30 meters. The shale is soft and cannot be cut into regular specimens. However, it breaks readily into small nuts of irregular and variable shape with linear dimensions of 1 to 5 cm, and these were oriented and measured directly. The results from each site are listed in Table 1.

The initial directions have negative inclinations with a tendency toward the present field direction. After thermal cleaning at 300° the directions at CS1 become more closely grouped, but at the remaining sites the scatter increases, this being particularly noticeable at CS4. However, the tendency toward the earth's field direction is lost, and the mean site directions are moved systematically away from this direction toward the west with steep negative inclination. The scatter of mean site directions appears to increase slightly, but this change is not significant. Although the within- and between-site precision is either little affected or is diminished by treatment, the tendency toward the present earth's field is lost, and for this reason the results after treatment are considered to be more reliable. The site mean directions are closely grouped ($N = 4$, $R = 3.97$, $k = 100$), and their mean is 338, −82. The error giving each site unit weight is 13°.

To check whether this high inclination also occurs at depth, a study was made of Chocolate shales in four bore cores obtained in connection with the construction of the Warragamba Dam. In this area, two shale horizons occur. Several

TABLE 1. Chocolate Shale Directions

The site numbering has no stratigraphical significance. S is the number of oriented samples. D_m and I_m give the direction of the resultant R. The directions are given relative to the present horizontal at each site, the beds being flat-lying or dipping by no more than a degree or so.

Site No.	Site Description	S	Natural Remanence D_m	Natural Remanence I_m	Natural Remanence R	After Heating to 300°C D_m	After Heating to 300°C I_m	After Heating to 300°C R
CS4	Roadside quarry Stanwell Tops (34°13′S, 150°59′E)	9	14	−76	8.47	23	−79	6 36
CS3	Road-cutting near Orford (34°13′S, 151°01′E)	9	32	−76	8 76	345	−82	8.13
CS2	Quarry Mt. Victoria (33°35′S, 150°14′E)	8	358	−75	7 86	352	−79	7.59
CS1	North shore of Long Reef (33°44′S, 151°19′E)	6	15	−70	5.77	269	−77	5.84

TABLE 2. Magnetic Inclination in Chocolate Shales in Bore Cores

The cores are from near Warragamba. The hole numbers are those used by the geological section of the Sydney Metropolitan Water Sewerage and Drainage Board, the cores being kindly made available by that organization. I_m is the average inclination relative to the bedding plane, which is horizontal in this region; the standard deviation (σ) and the 95 per cent error in the mean obtained using Gaussian statistics are also given.

Hole No.		Depth beneath Surface, meters	Thickness Covered, meters	No. of Core Samples	I_m	σ	Error ($P = 0.05$)
T214		53	1.2	17	−79	7	4
T212	Upper shale	72	1.5	15	−82	4	2
	Lower shale	87	0.3	5	−78	4	4
V1		154	0.8	14	−77	8	4
3 BT 50		86	1.2	11	−81	5	5

specimens were measured at each intersection of the shale horizons which range from 0.5 to 5 meters in thickness. All four bores intersect the upper shale horizon, but samples were available from only a single crossing of the lower shale horizon, which is about 15 meters below. The inclination but not the declination (since the azimuth of the cores was unknown) of the natural remanent magnetization was measured, and the results are summarized in Table 2. The inclinations are indistinguishable, as a group, from those obtained in surface outcrops after heating to 300°C, suggesting that the magnetization of the cores is more stable than in surface samples.

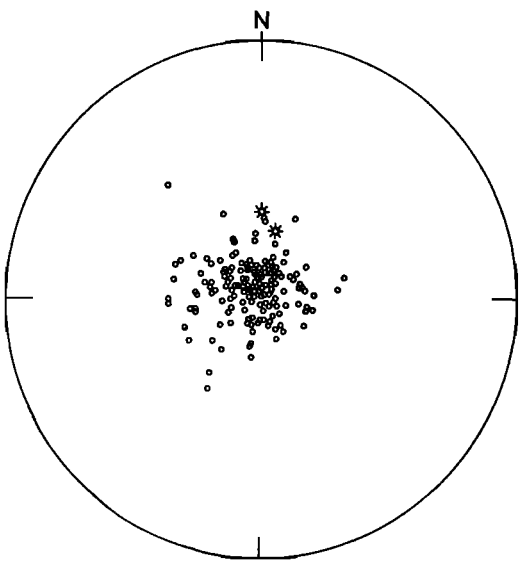


Fig. 1. Tasmanian dolerite directions. The primitive is the horizontal plane at the collecting localities. N points to true north. The directions (156 specimens) have negative inclinations and are plotted as circles on the upper hemisphere of an equal-area projection. The stars give the present field direction and that of the geocentric axial dipole field; these differ somewhat within the sampling area but not by more than the size of the markings.

Regional survey of the Tasmanian dolerites. Tasmanian dolerites are intrusive through folded Precambrian to Lower Devonian beds into several thousand meters of Permian and Triassic sediments, where they take the form of sills (generally 300 meters or more thick) and large dikelike bodies. The enclosing sediments often dip at small angles, but it is not certain whether this arose as a result of intrusion or of subsequent block tilting. No correction was applied to the directions. The intrusion occurred after the Triassic and before the Lower Tertiary period of faulting and sedimentation. The geological evidence therefore indicates a Jurassic or Cretaceous age, and by analogy with the Karroo dolerites of South Africa they have been considered to be Jurassic, possibly Lower Jurassic [*Banks*, 1958]. A more precise age has recently been given by *Evernden and Richards* [1961] and *McDougall* [1961] from radioisotope studies indicating a Lower to Middle Jurassic age (see discussion by *Stott* [1963] in the following paper).

Numerous inclination measurements on vertical bore cores have been made previously by other workers [*Jaeger and Joplin*, 1955; *Almond et al.*, 1956]. Typical values obtained by these workers are as follows: borehole 5001, 30 samples, average inclination 86°; borehole 5002,

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TABLE 3. Direction of Natural Remanent Magnetization in the Tasmanian Dolerites

S is the number of oriented samples and *N* the specimens cut from these. *D* and *I* are the declination and inclination of the resultant *R* of the specimen directions.

Site No.	Description	*S*	*N*	*R*	*D*	*I*
TD1	Quarry near Derwent Park Road, Hobart (42°50′S, 147°19′E)	2	2	1.97	63	−78
TD2	Quarry near Park Street, Hobart (42°52′S, 147°19′E)	2	2	1.97	258	−77
TD3	Cutting on Hobart to Sorell Road near Bellerive (42°52′S, 147°22′E)	2	2	1.95	234	−76
TD4	Quarry, Little Oyster Cove near Kettering (43°07′S, 147°16′E)	2	2	1.99(9)	54	−74
TD5	Road-cutting 2 miles west of Woodbridge on Gardiner Bay Road (43°10′S, 147°12′E)	2	2	1.91	279	−86
TD6	Road-cutting 1 mile north of Cradoc on Woodstock Road (43°06′S, 147°01′E)	2	2	1.99(8)	134	−85
TD7	Road-cutting on Woodstock-Huonville Road 1 mile south of site 8 (43°03′S, 147°02′E)	2	2	1.99(9)	281	−70
TD8	Quarry 1 mile southeast of Huonville near Woodstock road (43°02′S, 147°02′E)	2	2	1.99(9)	202	−81
TD9	Road-cutting at summit of Huon highway Neika (42°56′S, 147°14′E)	2	2	1.99(9)	294	−61
TD10	Quarry near Lyell highway, Gretna (42°42′S, 146°57′E)	2	2	1.99(8)	290	−76
TD11	Quarry near Lyell highway 1 mile east of Hamilton (42°33′S, 146°52′E)	2	2	1.99(6)	288	−72
TD12	Quarry near road between Ouse and Tarraleah (42°23′S, 146°35′E)	2	2	1.99	123	−78
TD13	Road-cuttings near Tarraleah power station (42°18′S, 146°28′E)	2	2	1.98	232	−86
TD14	Tunnel excavation near Tungatinah power station (42°17′S, 146°30′E)	2	2	1.99	356	−84
TD15	Road-cutting at Butler's Gorge beneath main dam (42°17′S, 146°17′E)	2	2	1.97	308	−80
TD16	Quarry 150 yards north of camp at Butler's Gorge (42°17′S, 146°17′E)	2	2	1.96	6	−85
TD17	Quarry 200 yards north of main dam at Butler's Gorge (42°16′S, 146°17′E)	2	2	1.98	76	−87
TD18	Quarry 3 miles north of Bronte (42°06′S, 146°30′E)	2	2	1.99(9)	24	−82
TD19	Quarry ¾ mile south of Kempton (42°33′S, 147°12′E)	2	2	1.97	330	−76
TD20	Quarry 2 miles north of Melton Mowbray (42°28′S, 147°13′E)	2	2	1.93	227	−76
TD21	Quarry near Bothwell (42°22′S, 147°00′E)	2	2	1.93	269	−75
TD22	Cutting along Liawenee canal (41°54′S, 146°40′E)	2	2	1.99(8)	257	−86
TD23	Quarry near Latrobe (41°15′S, 146°25′E)	2	2	1 95	16	−80
TD24	Quarry at Cataract Gorge, Launceston (41°27′S, 147°09′E)	5	8	7.84	16	−88
TD25	Road-cutting on Arthur highway 1½ miles southeast of Murdunna (42°57′S, 147°53′E)	2	2	1.99	357	−71
TD26	Road-cutting, Paradise Gorge, Orford (42°34′S, 147°51′E)	2	2	1.99	28	−78
TD27	Road-cutting near reservoir north of Triabunna (42°29′S, 147°55′E)	2	2	1.99	1	−80
TD28	Road-cutting on Tasman highway 2 miles south of Swansea (42°10′S, 148°02′E)	2	2	1.98	68	−81
TD29	Road-cutting east side of Swan River bridge, Cranbrook (42°01′S, 148°02′E)	2	2	1.99	30	−86
TD30	Road-cutting on Tasman highway between Bicheno and Cranbrook (41°59′S, 148°08′E)	2	2	1.99(8)	134	−87

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TABLE 3. Continued

Site No.	Description	S	N	R	D	I
TD31	Giblin Street, quarry, Hobart (42°52′S, 147°17′E)	1	2	1.97	324	−69
TD32	Quarry near Midland highway, 2 miles south of Oatlands (42°19′S, 147°20′E)	2	4	3.56	12	−75
TD33	Road-cutting on Huon highway near Longley (42°58′S, 147°12′E)	2	4	3.97	334	−79
TD34	Road-cutting, Police Point (43°15′S, 147°02′E)	2	4	3.93	228	−70
TD35	Road-cutting near Desolation Bay jetty (43°13′S, 147°00′E)	2	3	2.94	348	−83
TD36	Quarry 1½ miles north of Shipwright's Point (43°09′S, 146°59′E)	2	3	2.90	289	−79
TD37	Quarry 1½ miles south of Franklin (43°06′S, 147°00′E)	2	4	3.97	331	−87
TD38	Blackmans Bay sea cliffs (43°00′S, 147°19′E)	4	4	3.96	267	−71
(TD)T1	Road-cutting 3¾ miles west of Cygnet [*Robertson and Hastie*, 1962] (43°10′S, 147°00′E)	3	9	8.91	329	−83
(TD)T2	Road-cutting 4 miles southwest of Cygnet [*Robertson and Hastie*, 1962] (43°11′S, 147°00′E)	3	7	6.97	11	−85
TD39	Road-cutting, Mt. Wellington, 0–4 ft above contact, alt. 2350 ft (42°53′S, 147°14′E)	7	7	6.88	110	−79
TD40	Road-cutting, Mt. Wellington, alt. 2380 ft (42°53′S, 147°14′E)	4	4	3.98	317	−81
TD41	Road-cutting, Mt. Wellington, alt. 2420 ft (42°53′S, 147°14′E)	3	3	2.89	271	−83
TD42	Road-cutting, Mt. Wellington, alt. 2490 ft (42°53′S, 147°14′E)	4	4	3.85	324	−68
TD43	Road-cutting, Mt. Wellington, alt. 2530 ft (42°53′S, 147°14′E)	4	4	3.97	324	−70
TD44	Road-cutting, Mt. Wellington, alt. 2630 ft (42°53′S, 147°14′E)	4	4	3.99	323	−80
TD45	Road-cutting, Mt. Wellington, alt. 2650 ft (42°53′S, 147°14′E)	4	4	3.87	265	−81
TD46	Road-cutting, Mt. Wellington, alt. 2680 ft (42°53′S, 147°14′E)	4	4	3.90	318	−80
TD47	Road-cutting, Mt. Wellington, alt. 2800 ft (42°53′S, 147°14′E)	4	4	3.96	340	−86
TD48	Road-cutting, Mt. Wellington, alt. 3220 ft (42°53′S, 147°14′E)	4	4	3.93	54	−80
TD49	Road-cutting, Mt. Wellington, alt. 3430 ft (42°53′S, 147°14′E)	4	4	3.91	2	−78

7 samples, average inclination 84°; borehole 8102, 8 samples, average inclination 86°; borehole 8103, 9 samples, average inclination 86°. An account of a field survey, less complete than the present one, was reported by *Irving* [1956], and these results are incorporated here.

A total of 132 samples (156 specimens) were collected from 51 sites (TD149 and T1 and 2) covering much of the outcrop of these dolerites over an area of about 20,000 km². All specimen directions are plotted in Figure 1. Mean site directions are listed in Table 3. Results from two sites (T1 and T2) obtained by *Robertson and Hastie* [1962] are included.

The following field test suggests that the magnetic directions in the dolerite have remained unaltered since the Lower Tertiary. At Cartwright Point, south of Hobart, there is a fault scarp breccia of Lower Tertiary age (S. W. Carey, private communication) containing dolerite boulders with a highly irregular arrangement. If the magnetization of these dolerite boulders was imposed after formation of the breccia the directions should be parallel from boulder to boulder, but if the directions have remained unaltered they will be widely scattered. Directions from nine boulders are plotted in Figure 2, and it can be shown statistically that

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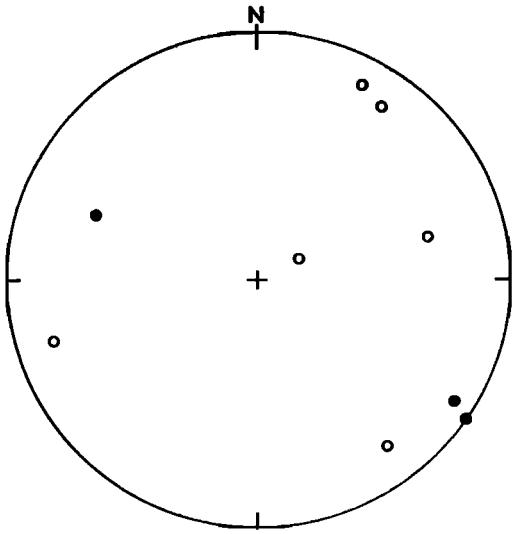


Fig. 2. Dolerite breccia directions. The primitive is the horizontal plane, dots indicating positive inclination and circles negative inclination. An equal-area projection is used.

these are random ($N = 9$, $R = 3.22$). This suggests that the directions of magnetization have remained unchanged since before the formation of this breccia. Although, as *Graham* [1961] suggests, it is possible that such scattered directions could arise from a lightning strike, this is considered unlikely in this case because at not one of the other 51 sites were similar scattered directions encountered (cf. Figures 1 and 2); moreover the intensities were similar to those observed in dolerite in situ. Further indications of stability are the good agreement in the directions observed between different igneous bodies both in surface exposures and in bore cores over a wide area and their systematic departure from the present field direction. Reference is made to the following paper [*Stott*, 1963], in which detailed laboratory studies of these dolerites are reported, and because of this demonstration of stability no systematic magnetic or thermal cleaning was necessary or was undertaken. All directions are negative (normal), although positive directions occur in volcanic tuffs adjacent to the dolerite. Positive directions observed in a dolerite bore core [*Jaeger and Joplin*, 1955] were not observed in a hole drilled 170 meters away [*Jaeger and Thyer*, 1960] and could have been due to systematic orientation errors.

The precision of directions is high, and the application of the two-tier analysis is justified. The site mean directions are well grouped ($N = 51$, $R = 50.05$), the average direction being 319, −84. The estimate of the within-site precision k_w is 41 and the between-site precision k_b is 111. The latter corresponds to a circular standard deviation of 8°, and it may be a measure of the secular variation at the time the dolerites cooled. The high precision of values is consistent with their high paleomagnetic latitude. The error is the mean ($P = 0.05$) is 3°.

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